

EXPERIMENT – 14

Study and Demonstration of Brute Force Password Attack in a Controlled Kali Linux Environment

Aim:

To study and demonstrate the concept of brute force password attacks in a controlled Kali Linux environment to understand password vulnerabilities and the importance of strong authentication mechanisms.

Software Requirements:

- **Host Operating System:** Windows / Linux / macOS
- **Virtualization Software:** Oracle VirtualBox
- **Guest Operating System:** Kali Linux
- **Study Tools:** Open-source password auditing utilities (educational use only)
- **System Requirements:**
 - RAM: Minimum 4 GB (8 GB recommended)
 - Disk Space: Minimum 30 GB
 - Processor: Intel/AMD with Virtualization support

Procedure:

Step 1: Start Kali Linux Virtual Machine

1. Open Oracle VirtualBox.
2. Start Kali Linux.
3. Log in using Kali credentials.

Step 2: Set Up a Controlled Test Environment

1. Use a **locally created test account** or **dummy authentication service**.
2. Ensure no real systems or credentials are involved.
3. Confirm the environment is isolated from external networks.

Step 3: Study Brute Force Attack Workflow

1. Understand how login mechanisms validate credentials.
2. Analyze how attackers automate repeated login attempts.
3. Observe how password complexity affects attack feasibility.

Step 4: Educational Demonstration (Simulation Only)

1. Simulate repeated authentication attempts using dummy data.
2. Observe system responses such as delays or access denial.
3. Record observations without executing real attacks.

Step 5: Analyze Security Weaknesses

- Weak or short passwords
- No rate limiting
- Missing account lockout policies
- Lack of multi-factor authentication

Step 6: Study Prevention and Mitigation Techniques

- Strong password policies
- Account lockout and rate limiting
- CAPTCHA mechanisms
- Multi-factor authentication (MFA)
- Monitoring and alerting

Output:

```
import time

# A mock database of user credentials
USER_DB = {
    "admin": "password123"
}

def authenticate(username, password):
    """Simulates a login mechanism validating credentials."""
    # Simulating network/processing delay
    time.sleep(0.1)

    if username in USER_DB and USER_DB[username] == password:
        return True
    return False
```

```
from dummy import authenticate

# Dummy data: A short dictionary of common weak passwords
password_list = ["123456", "admin", "qwerty", "letmein", "password123", "admin123"]
target_user = "admin"

print(f"Starting brute force simulation against user: {target_user} ... \n")

attempts = 0
for password in password_list:
    attempts += 1
    print(f"Attempt {attempts}: Trying password '{password}' ... ")

    # Observe system responses such as access denial
    if authenticate(target_user, password):
        print(f"\n[+] SUCCESS! Password found: {password}")
        break
    else:
        print("[-] Access Denied.")
```

```
(kali@kali)~$ python3 brute_force.py
Starting brute force simulation against user: admin...

Attempt 1: Trying password '123456' ...
[-] Access Denied.
Attempt 2: Trying password 'admin' ...
[-] Access Denied.
Attempt 3: Trying password 'qwerty' ...
[-] Access Denied.
Attempt 4: Trying password 'letmein' ...
[-] Access Denied.
Attempt 5: Trying password 'password123' ...
[+] SUCCESS! Password found: password123
```

Result:

The study and demonstration of brute force password attacks were successfully completed in a controlled Kali Linux environment.